

Лабораторная работа №1

Основы LaTeX, компиляция и специальные символы

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Цель работы

- Изучение базовой структуры LaTeX-документа
- Освоение процесса компиляции `pdflatex`
- Изучение преамбулы, окружений, специальных символов
- Создание и тестирование первого документа LaTeX

Выполнение работы

Установка TeX Live через Chocolatey

- Выполнена команда: `choco install texlive`
- Система скачала и установила множество пакетов
- Завершение установки прошло успешно

```
Administrateur : Windows PowerShell

PS C:\WINDOWS\system32> Test-Path "C:\ProgramData\chocolatey\bin"
True
PS C:\WINDOWS\system32> choco install texlive
Chocolatey v2.5.1
Installing the following packages:
texlive
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading texlive 2025.20250902.0... 100%

texlive v2025.20250902.0 [Approved]
texlive package files install completed. Performing other installation steps.
The package texlive wants to run 'chocolateyinstall.ps1'.
Note: If you don't run this script, the installation will fail.
Note: To confirm automatically next time, use '-y' or consider:
choco feature enable -n allowGlobalConfirmation
Do you want to run the script?([Y]es/[A]ll scripts/[N]o/[P]rint): y

Additional parameters for installer:
Writing Profile
Writing Profile using Passed Parameters.
TeX Live Profile will be written to C:\tools\texlive.profile
Extracting C:\ProgramData\chocolatey\lib\texlive\tools\install-tl.zip to C:\tools\install-tl-27092518...
C:\tools\install-tl-27092518
Automated TeX Live installation using profile: C:\tools\texlive.profile
Loading https://mirror.truenetwork.ru/CTAN/systems/texlive/tlnet/tlpkg/texlive.tlpdb
Installing TeX Live 2025 from: https://mirror.truenetwork.ru/CTAN/systems/texlive/tlnet (not verified)
Platform: windows => 'Windows (64-bit)'
Distribution: net (downloading)
Using URL: https://mirror.truenetwork.ru/CTAN/systems/texlive/tlnet
Directory for temporary files: C:\Users\ousma\AppData\Local\Temp\TX0cyii2bT
Installing to: C:\texlive\2025
Installing [1/6, time/total: ????:??:?]: hyphen-base [23k]
Installing [2/6, time/total: 00:02/24:23]: kpathsea [1042k]
Installing [3/6, time/total: 00:04/01:00]: luatex [1957k]
Installing [4/6, time/total: 00:07/00:37]: texlive-scripts [556k]
Installing [5/6, time/total: 00:09/00:40]: texlive.infra [610k]
Installing [6/6, time/total: 00:12/00:50]: tlmgrl.window [11801k]
```

- Установщик сообщил об успешной конфигурации
- Переменные окружения обновятся после перезапуска PowerShell

```
pre-generating all format files, be patient...
[18:08:17] running fntutil-sys --no-error-if-no-engine-luametatex, luajitbtx, luajitx, mfluajit --no-strict --all ...done
running path adjustment actions
Finished with path adjustment actions
[18:08:44] running package-specific postactions
[18:08:45] finished with package-specific postactions

logfile: C:\texlive\2025\install-tl.log
Environment Vars (like PATH) have changed. Close/reopen your shell to
see the changes (or in powershell/cmd.exe just type 'refreshenv').
The install of texlive was successful.
  Deployed to 'C:\tools\install-tl-27092518'

chocolatey installed 1/1 packages.
See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\WINDOWS\system32> refreshenv
RefreshEnv.cmd does not work when run from this process. If you're in PowerShell, please 'Import-Module $env:ChocolateyInstall\helpers\chocolateyProfile.ps1' and try again.
PS C:\WINDOWS\system32>
```

Рис. 2: Завершение установки TeX Live

Компиляция тестовых файлов

- Проверена работа **pdflatex** на учебных документах
- Система корректно сгенерировала **.pdf**
- Созданы вспомогательные файлы (**.log**, **.aux**, **.pdf**)

```
Administrateur : Windows PowerShell
Ok [25] [26] [28] [27] [29] [30] [31])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/excomp.mf
Ok [38] [63] [190])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exrullett.mf
Ok [65] [66] [67] [68] [69] [70] [71] [72] [73] [74] [75] [76] [77] [78]
[79] [80] [81] [82] [83] [84] [85] [86] [87] [88] [89] [90])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exrllett.mf
Ok [97] [98] [99] [100] [101] [102] [103] [104] [105] [106] [107] [108]
[109] [110] [111] [112] [113] [114] [115] [116] [117] [118] [119] [120]
[121] [122]) (c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exrdigit.mf
Ok [48] [49] [50] [51] [52] [53] [54] [55] [56] [57])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exrligb.mf
Ok) ) )
(some charht values had to be adjusted by as much as 0.13197pt)
Font metrics written on ecbx1728.tfm.
Output written on ecbx1728.600gf (256 characters, 84848 bytes).
Transcript written on ecbx1728.log.
gftopk.exe ecbx1728.600gf ecbx1728.600pk
<c:/Users/ousma/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecbx1728.600pk>
Output written on step17-custom.pdf (1 page, 9960 bytes).
Transcript written on step17-custom.log.
PS C:\LaTeX-Projet> # Vérifier tous les fichiers créés
>> dir step*.tex
>> dir step*.pdf
```

Répertoire : C:\LaTeX-Projet

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	27/09/2025 19:39	254	step10-citations.tex
-a----	27/09/2025 19:49	474	step11-advanced.tex
-a----	27/09/2025 20:29	281	step13-classes.tex
-a----	27/09/2025 20:30	286	step14-packages.tex
-a----	27/09/2025 20:31	524	step15-complete.tex
-a----	27/09/2025 20:31	155	step16-errors.tex
-a----	27/09/2025 20:32	182	step17-custom.tex

```
\documentclass{article}
\usepackage[T1]{fontenc}

\begin{document}
Hey world!
This is a first document.
\end{document}
```



```
Hey world!  
This is a first document.
```

Отработаны команды вывода:

- короткие команды типа `\%`, `\{`, `\_`
- длинные команды: `\textbackslash`, `\textasci...`
- использование жёсткого пробела `~`

1 Special characters test

Testing special characters from the book:

Symbol	Short Command	Long Command
{	\{	\textbraceleft
}	\}	\textbraceright
\$	\\$	\textdollar
%	\%	
&	\&	
#	\#	
_	_	\textunderscore
\		\textbackslash
^		\textasciicircum
~		\textasciitilde

Hard spaces: Mr. Smith, page 45, Fig. 1.

Итоги работы

В ходе работы были освоены:

- структура LaTeX-документа: преамбула, тело, окружения
- процесс компиляции `pdflatex`
- создание минимального документа
- работа со специальными символами
- тестирование корректности установки TeX Live

Система LaTeX функционирует корректно, компиляция проходит успешно.